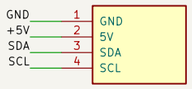
HastaDAQ strain-gauge board

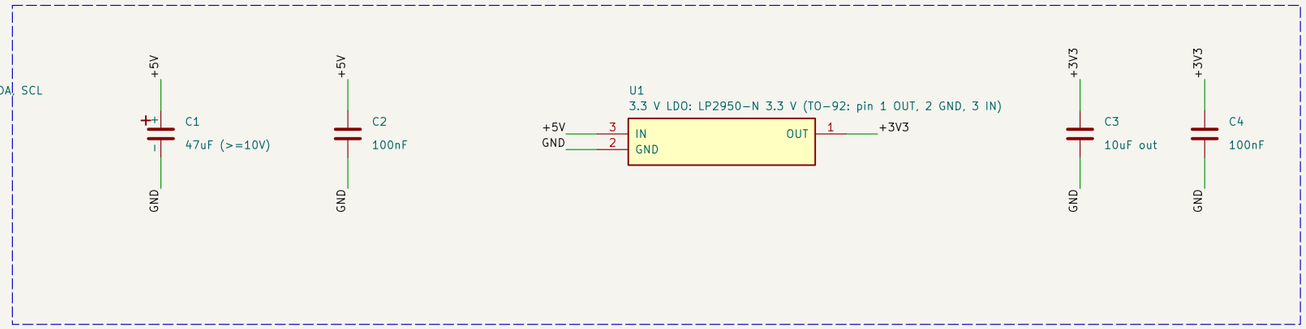
Net labels connect everything: same name = same net. Companion to the main board (kicad/hastadaq). Reusable: any I2C master with 5 V and 3.3 V logic can drive it.

J1
LOOM from main board (4-wire I2C): GND, +5V, SDA, SCL



Pin 2 is +5V, NOT 3V3: use a keyed plug, never a Qwiic cable.

POWER: bulk + decoupling on the loom 5 V (any ferrite bead is part of the loom, not the board), the LDO makes the quiet 3.3 V that powers BOTH ADCs and the bridge excitation



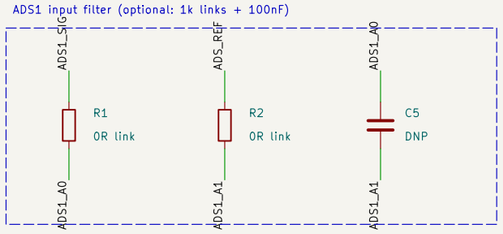
Load is about 20 mA (bridges 2 x 4.7 mA + ADCs): the LDO dissipates about 35 mW.



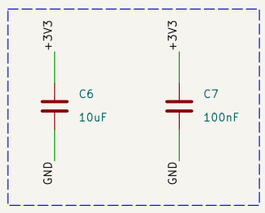
U2
Adafruit ADS122C04 #1 (I2C 0x40)



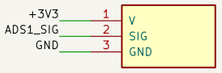
ADS1 input filter (optional: 1k links + 100nF)



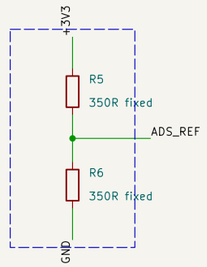
ADS1 decoupling (at the ADC)



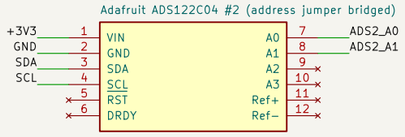
J2
HB1 half-bridge #1 (on beam): V, SIG, GND



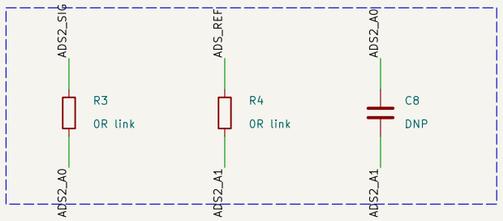
Shared completion resistors (fixed) -> A1 of BOTH ADCs



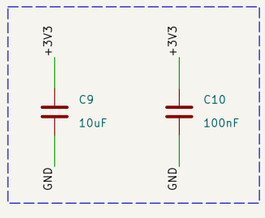
U3
Adafruit ADS122C04 #2 (address jumper bridged)



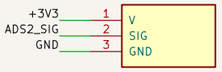
ADS2 input filter (optional: 1k links + 100nF)



ADS2 decoupling (at the ADC)



J3
HB2 half-bridge #2 (on beam): V, SIG, GND



NOTES
1. Each half-bridge head (2 gauges) plugs into a 3-pin header: V (+3V3), SIG (mid-point), GND. The two fixed 350R completion resistors are shared; their mid-point (ADS_REF) goes to A1 of both ADCs. Excitation is the ADC's own VIN, so VREF_SUPPLY is ratiometric: no Ref+/Ref- wiring.
2. 3V3 comes ONLY from the on-board LDO (from the loom's 5 V): cable voltage drop and the main board's switching noise do not reach the bridges or the ADC reference. The ADC logic lines (SDA/SCL) are pulled up by the ADS122C04 breakouts to this 3V3 and are 3.3 V logic: the master must not drive them above 3.6 V.
3. Input filters are optional: with the 0R links and no capacitor fitted there is none (original behaviour). Fit 1k in each link and 100nF across A0-A1 for about an 800 Hz low-pass.
4. ADS #2: bridge one address jumper on its back (resulting address unconfirmed; the main board's firmware scans at boot).
5. Use shielded twisted cable for the gauge wires, shield to GND at this board only.

HastaArduino		
Sheet: /		
File: gaugeboard.kicad_sch		
Title: HastaDAQ strain-gauge board		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1